Visualizing selection analysis results for evolution of the BA.2.75.2 (WHO Omicron) clade

Display this analysis.

BA.2.75.2 (all sequences as of 09.18.2022)

Loaded data on 22 segments and 9267 sites

Objective.

We investigated the nature and extent of selective forces acting on the viral genes in *BA.2.75.2* clade sequences by performing a series of comparative phylogenetic analyses on a median of 24 *BA.2.75.2* sequences and a median of 141 sequences from available sequences chosen to represent genomic diversity in SARS-CoV-2.

Results.

1. There is evidence of **diversifying positive selection** acting on the *BA.2.75.2* sequences, on **46** individual sites (there are 356 sites that are polymorphic in the amino-acid space among clade sequences; selection also includes the basal branch of the clade) along the internal branches of the clade (Table 1).
2. There is evidence of **diversifying positive selection** acting on the *BA.2.75.2* sequences, on **38** individual sites along **all** branches of the clade (Table 7).
3. When comparing the strength of selection on *BA.2.75.2* to reference sequences along internal tree branches, **16** individual sites along the internal branches of the clade (Table 2) showed statistically significant differences.
4. Over the entire tree **0** sites show evidence of directional selection (Table 3).
5. Along the internal branches of the *BA.2.75.2* clade, **0** pairs of sites showed evidence of **coevolution** (Table 4).
6. Along the internal branches of the *BA.2.75.2* clade, **8** sites showed evidence of **negative selection** (Table 5).
7. Along the internal branches of the *BA.2.75.2* clade, **1** genes/ORFs showed evidence of **episodic diversifying selection** (BUSTED, q-value ≤ 0.1, Table 6).

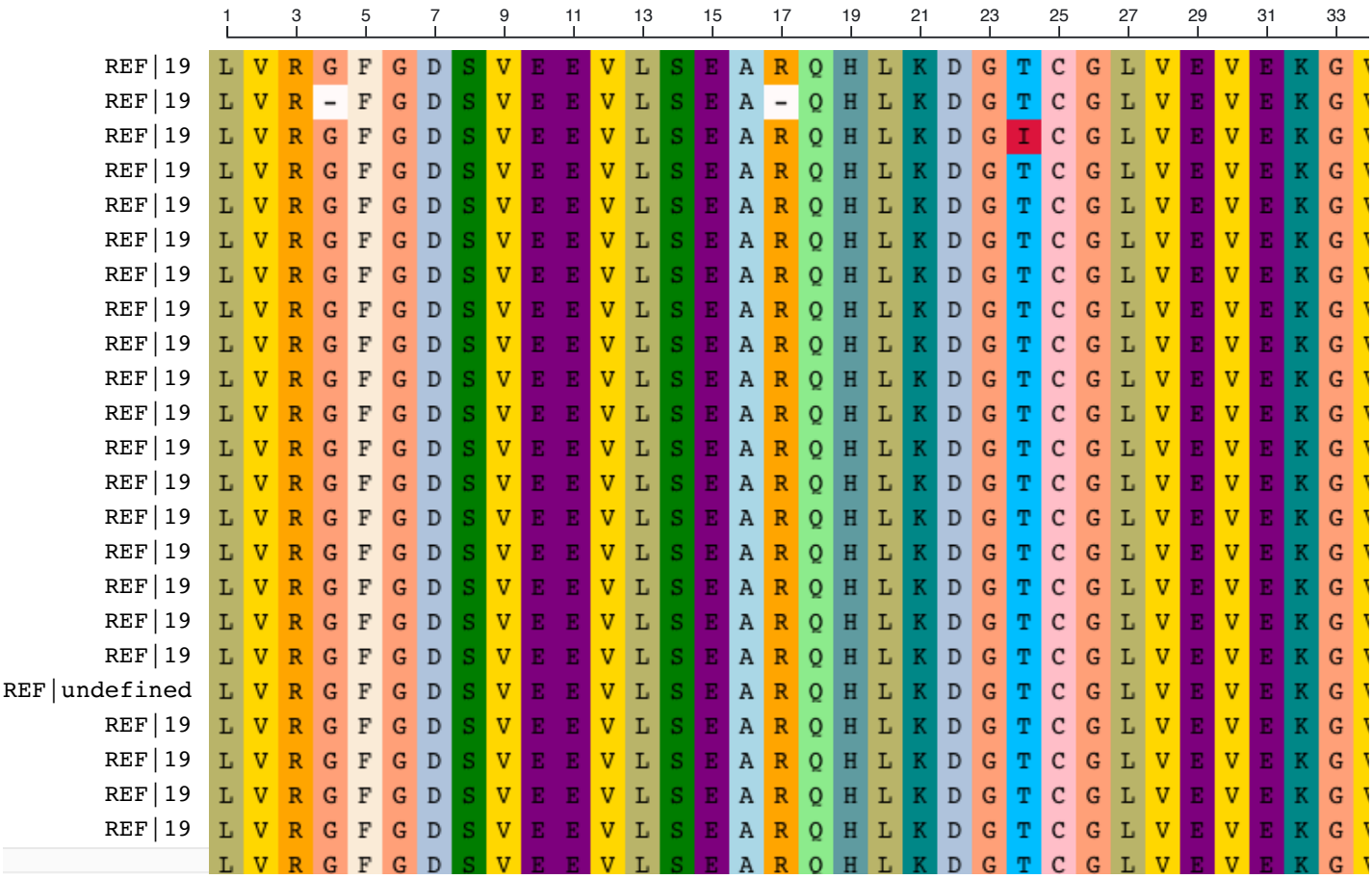
Segment information

View this genomic segment

leader

The BUSTED p-value for episodic selection is **0.500**. The RELAX p-value is **0.0109** and the relaxation/intensification parameters was **0.417**, implying that selection in BA.2.75.2 was **relaxed** relative to the reference branches.

Figure 0. Segment alignment view



Site-level information

View this site (in SARS-CoV-2 reference coordinates)

Select a site (or overall)

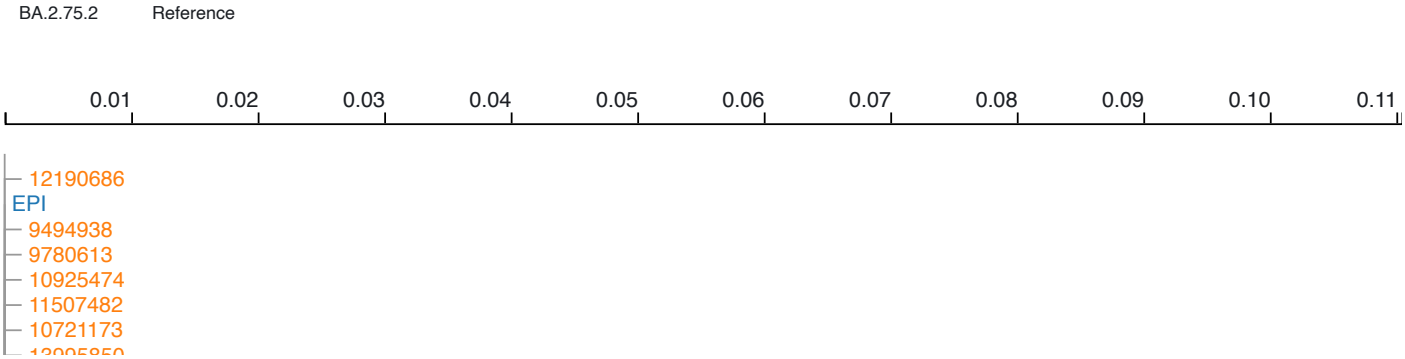
Select a site to view detailed information on it

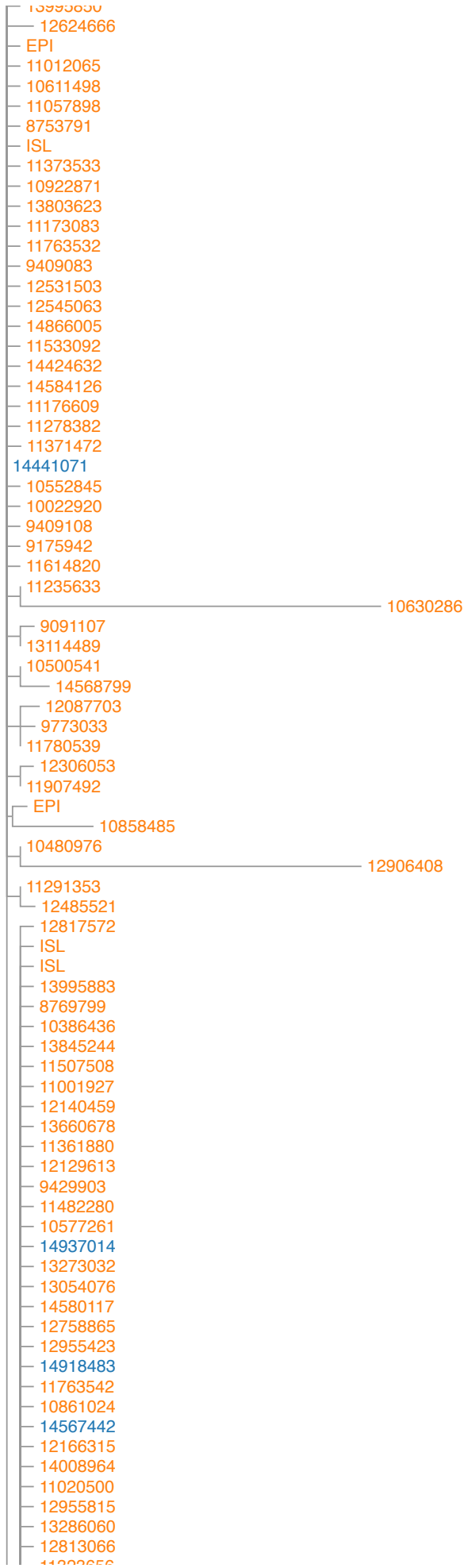
1024 x 800 Resize

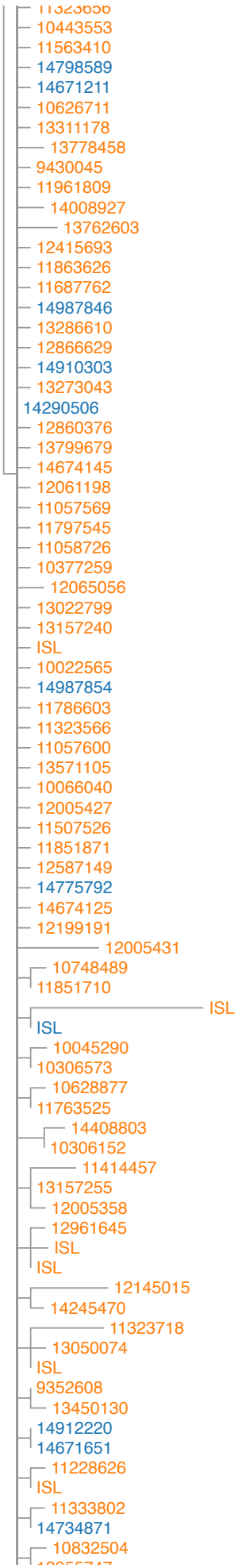
Tree dimension (height x width in pixels), leave blank to auto-scale

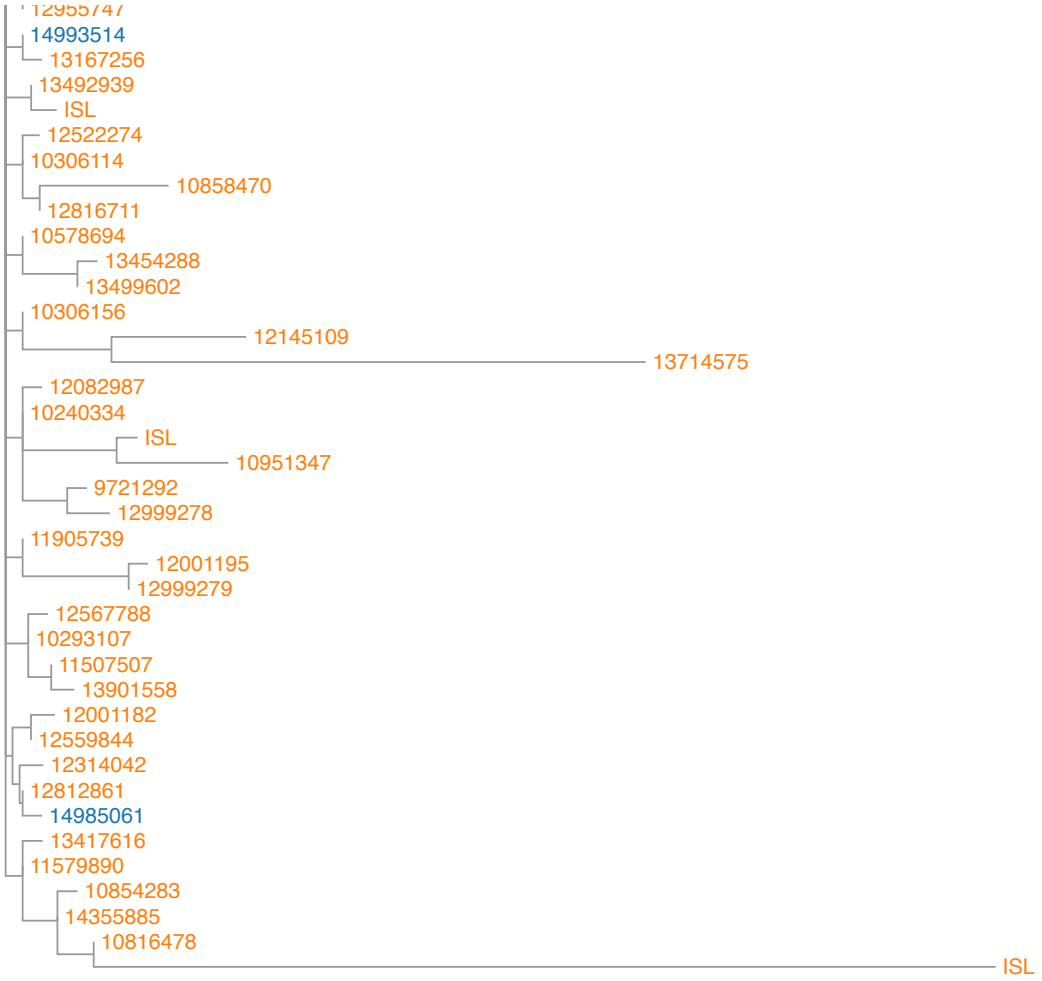
☐ Label tree with amino-acids

Branches that have experienced at least one substittuion are shown in thicker lines. Branches that have MEME-based support for episodic selection at a given site, are shown in **dark red**. Mouse over branches to show additional information if available (e.g substitutions over those branches, and MEME episodic selection support).




13995850
12624666
EPI
11012065
10611498
11057898
8753791
ISL
11373533
10922871
13803623
11173083
11763532
9409083
12531503
12545063
14866005
11533092
14424632
14584126
11176609
11278382
11371472
14441071
10552845
10022920
9409108
9175942
11614820
11235633
10630286
9091107
13114489
10500541
14568799
12087703
9773033
11780539
12306053
11907492
EPI
10858485
10480976
12906408
11291353
12485521
12817572
ISL
ISL
13995883
8769799
10386436
13845244
11507508
11001927
12140459
13660678
11361880
12129613
9429903
11482280
10577261
14937014
13273032
13054076
14580117
12758865
12955423
14918483
11763542
10861024
14567442
12166315
14008964
11020500
12955815
13286060
12813066


11323656
10443553
11563410
14798589
14671211
10626711
13311178
13778458
9430045
11961809
14008927
13762603
12415693
11863626
11687762
14987846
13286610
12866629
14910303
13273043
14290506
12860376
13799679
14674145
12061198
11057569
11797545
11058726
10377259
12065056
13022799
13157240
ISL
10022565
14987854
11786603
11323566
11057600
13571105
10066040
12005427
11507526
11851871
12587149
14775792
14674125
12199191
12005431
10748489
11851710
ISL
ISL
10045290
10306573
10628877
11763525
14408803
10306152
11414457
13157255
12005358
12961645
ISL
ISL
12145015
14245470
11323718
13050074
ISL
9352608
13450130
14912220
14671651
11228626
ISL
11333802
14734871
10832504



Result tables

Table 1 List of sites found to be under diversifying positive selection by MEME ($p \leq 0.05$) along internal branches in *BA.2.75.2*, as well as biochemical properties that are *important* at this site (via the PRIME method)

Coordinate (SARS-CoV-2)	Gene/ORF	Codon (in gene/ORF)	# of selected branches	p-value	q-value	Properties
26274	E	11	1	0.000523	0.0118	
21640	S	27	3	0.0000591	0.00355	bipolar (constrained) , composition (changing), charge (changing)
25494	ORF3a	35	1	0.000308	0.00924	
532	ORF1a	90	1	0.000128	0.00574	
22000	S	147	1	0.0317	0.154	bipolar (constrained) , charge (changing), undefined (changing)
25842	ORF3a	151	1	0.000191	0.00688	
22030	S	157	0	0.0350	0.162	bipolar (constrained) , composition (changing), charge (changing), undefined (changing)
22189	S	210	2	0.0106	0.100	bipolar (constrained) , secondary (constrained) , undefined (changing)

940	ORF1a	226	1	0.0461	0.184
943	ORF1a	227	1	0.0259	0.146

1. **Coordinate (SARS-CoV-2)** : the starting coordinate of the codon in the NCBI reference SARS-CoV-2 genome
2. **Gene/ORF** : which gene or ORF does this site belong to
3. **Codon (in gene/ORF)** : the location of the codon in the corresponding Gene/ORF
4. **% of branches with ω>1** : the fraction of tree branches (internal branches in the BA.2.75.2 clade) that have evidence of diversifying positive selection at this site (100% -- pervasive selection, <100% -- episodic selection).
5. **p-value** : the p-value for the likelihood ratio test that a non-zero fraction of branches have ω > 1 (i.e. episodic diversifying selection at this branch). This is not corrected for multiple testing; the MEME test is generally conservative on real data.
6. **q-value** : multiple-test corrected q-value (Benjamini-Hochberg)
7. **Properties** : which, if any, of the five compositive biochemical properties from Atchley et al are conserved or changed at this site.

Table 2 List of sites found to be selected differentially along internal branches between *BA.2.75.2* and reference sequences (FDR ≤ 0.2) using the Contrast-FEL method

Coordinate (SARS-CoV-2)	Gene/ORF	Codon (in gene/ORF)	Ratio of ω (BA.2.75.2 : reference)	p	q
1375	ORF1a	371	NaN	0.0359	1.00
2209	ORF1a	649	NaN	0.0346	1.00
2776	ORF1a	838	NaN	0.0167	1.00
8215	ORF1a	2651	NaN	0.0174	1.00
13456	ORF1a	4398	NaN	0.0262	1.00
14257	ORF1b	264	NaN	0.0261	1.00
15706	ORF1b	747	NaN	0.0239	1.00
18927	ORF1b	1820	NaN	0.0123	1.00
21574	S	5	NaN	0.0000547	0.0697
22534	S	325	NaN	0.000493	0.314
23563	S	668	NaN	0.0158	1.00
27687	ORF7a	99	NaN	0.0487	1.00
27974	ORF8	28	NaN	0.0284	1.00
28112	ORF8	74	NaN	0.0478	1.00
28720	N	150	NaN	0.0490	1.00

1. **Coordinate (SARS-CoV-2)** : the starting coordinate of the codon in the NCBI reference SARS-CoV-2 genome
2. **Gene/ORF** : which gene or ORF does this site belong to
3. **Codon (in gene/ORF)** : the location of the codon in the corresponding Gene/ORF
4. **Ratio of ω (BA.2.75.2 : reference)** : the ratio of site-level ω estimates for the two sets of branches. If this ratio is >1, then selection on BA.2.75.2 is stronger. These values are highly imprecise and should be viewed as qualitative measures.
5. **p-value** : the p-value for the likelihood ratio test that ω ratios between the internal branches of the two clades are different. This is not corrected for multiple testing.
6. **q-value** : multiple-test corrected q-value (Benjamini-Hochberg)

Table 3 List of sites found to be evolving under directional selection in the entire tree, using a FUBAR-like implementation of the DEPS method. The tree was rooted on the

genome reference sequence for this analysis.

```
TypeError: Cannot convert undefined or null to object
```

1. **Coordinate (SARS-CoV-2)** : the starting coordinate of the codon in the NCBI reference SARS-CoV-2 genome
2. **Gene/ORF** : which gene or ORF does this site belong to
3. **Codon (in gene/ORF)** : the location of the codon in the corresponding Gene/ORF
4. **Target amino-acid** : which amino-acids have statistical support (Bayes Factor ≥ 100) for accelerated evolution towards them.

Table 4 Pairs of sites found to have epistatic (co-evolving) substitution patterns by BGM method

```
TypeError: Cannot convert undefined or null to object
```

1. **Gene/ORF** : which gene or ORF does this site belong to
2. **Codon 1/2 (in gene/ORF)** : the location of the two interacting codons in the corresponding Gene/ORF
3. **Posterior probability of non-independence** : estimated posterior probability that substitutions which occur on the interior branches of the BA.2.75.2 clade are **not** independent.

Table 5 List of sites found to be under pervasive negative selection by FEL ($p \leq 0.05$) along internal branches in *BA.2.75.2*

Coordinate (SARS-CoV-2)	Gene/ORF	Codon (in gene/ORF)	Synonymous rate	Non-synonymous rate	p-value	q-value
26522	M	1	614	0.00	0.0468	1.00
29503	N	411	1.00e+4	0.00	0.00624	0.561
15241	ORF1b	592	1.00e+4	0.610	0.0201	1.00
15442	ORF1b	659	1.00e+4	0.00	0.0239	1.00
6550	ORF1a	2096	1.00e+4	0.00	0.000169	0.0303
7024	ORF1a	2254	1.00e+4	0.00	0.0373	1.00
9457	ORF1a	3065	1.00e+4	0.00	0.0464	1.00
11731	ORF1a	3823	584	0.00	0.0395	1.0

1. **Coordinate (SARS-CoV-2)** : the starting coordinate of the codon in the NCBI reference SARS-CoV-2 genome
2. **Gene/ORF** : which gene or ORF does this site belong to
3. **Codon (in gene/ORF)** : the location of the codon in the corresponding Gene/ORF
4. **Synonymous rate** : Site estimate for the synonymous substitution rate (α). These values are highly imprecise and should be viewed as qualitative measures.
5. **Non-synonymous rate** : Site estimate for the non-synonymous substitution rate (β). These values are highly imprecise and should be viewed as qualitative measures.
6. **p-value** : the p-value for the likelihood ratio test that β/&alpha ≠ 1
7. **q-value** : multiple-test corrected q-value (Benjamini-Hochberg)

Table 6 BUSTED selection results on the SARS-CoV-2 clade across segments

Segment	ω1	p1	ω2	p2	ω3	p3	p-value	q-value

RdRp	0.00	99.4%	0.00	0.306%	166	0.276%	**0.000185**	**0.00407**
S	0.00915	41.9%	0.0115	51.4%	15.9	6.64%	**0.0255**	0.281
nsp6	0.00	11.5%	0.00	86.3%	27.4	2.26%	0.0609	0.447
nsp10	0.997	0.00%	1.00	0.0974%	25.9	99.9%	0.127	0.698
E	0.998	0.00%	1.00	0.0310%	151	100%	0.129	0.567
ORF8	0.00	78.9%	0.00	15.5%	8.79	5.59%	0.261	0.958
methyltransferase	0.00	17.3%	1.00	75.8%	26.1	6.83%	0.268	0.841
3C	0.299	93.7%	0.374	4.34%	1.00	1.95%	0.500	1.38
endornase	0.00709	87.9%	0.0917	3.82%	1.00	8.25%	0.500	1.22
exonuclease	0.00209	0.00%	0.207	100%	1.00	0.00%	0.500	1.10
helicase	0.0234	7.05%	0.0907	62.6%	1.00	30.3%	0.500	1.00
leader	0.243	88.1%	0.956	11.9%	1.81	0.00%	0.500	0.917
M	0.341	97.1%	0.742	2.92%	1.00	0.00%	0.500	0.846
N	0.0143	0.00%	0.336	100%	1.04	0.00%	0.500	0.786
nsp2	0.702	32.8%	0.723	51.0%	1.36	16.2%	0.500	0.733
nsp3	0.00144	0.00%	0.270	100%	1.39	0.00%	0.500	0.688

Table 7 List of sites found to be under diversifying positive selection by MEME ($p \leq 0.05$) along **all** branches in *BA.2.75.2*, as well as biochemical properties that are *important* at this site (via the PRIME method)

Coordinate (SARS-CoV-2)	Gene/ORF	Codon (in gene/ORF)	# of selected branches	p-value	p-value/int	q-value	Properties
21574	S	5	5	0.000122	0.667	0.00550	overall, bipolar (constrained)
26274	E	11	1	0.00227	0.000523	0.0372	
26292	E	17	1	0.00283	0.667	0.0364	
26298	E	19	1	0.000884	0.667	0.0199	
21640	S	27	4	0.00118	0.0000591	0.0212	bipolar (constrained) , composition (changing), charge (changing)
27588	ORF7a	66	1	0.0374	0.667	0.204	
26445	E	68	1	0.00103	0.667	0.0206	
26747	M	76	1	0.00586	0.667	0.0620	
22000	S	147	2	0.0464	0.0317	0.220	bipolar (constrained) , charge (changing), undefined (changing)
22189	S	210	3	0.0198	0.0106	0.155	bipolar (constrained) , secondary (constraine , undefined (changing)

1. **Coordinate (SARS-CoV-2)** : the starting coordinate of the codon in the NCBI reference SARS-CoV-2 genome
2. **Gene/ORF** : which gene or ORF does this site belong to
3. **Codon (in gene/ORF)** : the location of the codon in the corresponding Gene/ORF
4. **% of branches with ω>1** : the fraction of tree branches (internal branches in the BA.2.75.2 clade) that have evidence of diversifying positive selection at this site (100% -- pervasive selection, <100% -- episodic selection).
5. **p-value** : the p-value for the likelihood ratio test that a non-zero fraction of branches have $\omega > 1$ (i.e. episodic diversifying selection at this branch). This is not corrected for multiple testing; the MEME test is generally conservative on real data.
6. **p-value/int** : the p-value for the likelihood ratio test that a non-zero fraction of **internal** branches

6. **p-value/int** : the p-value for the likelihood ratio test that a non-zero fraction of **internal** branches have ω > 1 (i.e. episodic diversifying selection at this branch). This is not corrected for multiple testing; the MEME test is generally conservative on real data.
7. **q-value** : multiple-test corrected q-value (Benjamini-Hochberg)
8. **Properties** : which, if any, of the five compositive biochemical properties from Atchley et al are conserved or changed at this site.

Data and libraries

```
d3 = ▸ Object {format: ƒ(t), formatPrefix: ƒ(t, n), timeFormat: ƒ(t), timeParse: ƒ(t), utcForm

phylotree = ▸ Object {center_of_tree: ƒ(t, e), cluster_picker: ƒ(t, e, s, i, a), compute_midpo

_ = ƒ(value)

<link>

site2view = ""

tree_object = ▸ Array(2) [tt, Object]

▸ Q {css_classes: Object, phylotree: tt, container: "#tree_container_0", separation: ƒ(t, e),

import {table} from "@gampleman/table"

import {select, checkbox, autoSelect, text} from "@jashkenas/inputs"

vegalite = ƒ(…)

selection_profile = ▸ Array(176) [Object, Object, Object, Object, Object, Object, Object, Obje

fragments = ▸ Array(22) [Array(8), Array(8), Array(8), Array(8), Array(8), Array(8), Array(8),

float_format = ƒ(t)

cladeName = "BA.2.75.2"

plotPie = ƒ(data)

prime_annotation = ▸ Array(5) ["bipolar", "secondary", "volume", "composition", "charge"]

report_ratio = ƒ(a, b)

species = ▸ Array(2) ["BA.2.75.2", "Reference"]
```

```
speciesCount = ▸ Array(22) [Object, Object, Object, Object, Object, Object, Object, Object, Ob
species_colors = ƒ(i)
busted_distributions = ▸ Array(66) [Object, Object, Object, Object, Object, Object, Object, Ob
relax_distributions = ▸ Array(132) [Object, Object, Object, Object, Object, Object, Object, Ob
busted_distribution_summary = ▸ Array(22) [Object, Object, Object, Object, Object, Object, Obj
conserved_sites = ▸ Array(67) [Array(2), Array(2), Array(2), Array(2), Array(2), Array(2), Arr
site2segment = ▸ Object {265: "leader", 268: "leader", 271: "leader", 274: "leader", 277: "lea
site2alignment = ▸ Object {265: 1, 268: 2, 271: 3, 274: 4, 277: 5, 280: 6, 283: 7, 286: 8, 289
p_cut = 0.05
selected_sites = ▸ Array(46) [Object, Object, Object, Object, Object, Object, Object, Object,
selected_sites_leaves = ▸ Array(38) [Object, Object, Object, Object, Object, Object, Object, O
negative_sites = ▸ Array(8) [Object, Object, Object, Object, Object, Object, Object, Object]
diff_sites = ▸ Array(16) [Array(9), Array(9), Array(9), Array(9), Array(9), Array(9), Array(9)
directional_sites = ▸ Array(0) []
cfel_sites = ▸ Array(8) [Object, Object, Object, Object, Object, Object, Object, Object]
site_report = undefined
freq_report = ƒ(freqs)
cfel_site_report = ƒ(d)
fade_site_report = ƒ(d)
prime_site_report = ƒ(d)
coevolve_sites = ▸ Array(0) []
subsByBranch = ▸ Object {Node248: Array(180), root: Array(180), hCoV_19_India_RJ_SMS_ICMR_INSA
polymorphic_sites = ▸ Array(356) [Array(2), Array(2), Array(2), Array(2), Array(2), Array(2),
entropy = ƒ(array)
```

```
gene_coordinates = ▸ Array(1) [Array(4)]

name_fragment = ƒ(n)

summary_json = ▸ Object {3C: Object, E: Object, endornase: Object, exonuclease: Object, helica

has_full_meme = true

annotation_json = ▸ Object {265: Object, 268: Object, 271: Object, 274: Object, 277: Object, 2

import {legend, swatches} from "@d3/color-legend"

import {translation_table} from "@spond/usual-vs-unusual-mutation-analyses"

import {a, r} from "@stephenshank/alignment-js-in-obervablehq"

React = ▸ Object {Children: Object, Component: ƒ(a, b, c), Fragment: Symbol(react.fragment), P

ReactDOM = ▸ Object {__SECRET_INTERNALS_DO_NOT_USE_OR_YOU_WILL_BE_FIRED: Object, createPortal:

translation_table_gap = ▸ Object {TTT: "F", TCT: "S", TAT: "Y", TGT: "C", TTC: "F", TCC: "S",

translate_ambiguous_codon_gap = ƒ(c)

undefined

<link>
```